

Momentum distributions in $t\bar{t}$ production and decay near threshold[★]

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Abstract. The Green function formalism for the description of $t\bar{t}$ production in e^+e^- annihilation close to threshold is extended from the calculation of the total cross section to the prediction of momentum distributions of top quarks. The Lippmann-Schwinger equation is solved numerically in momentum space for the two loop QCD potential. The spectra of W bosons from top quark decays are calculated and found to be strongly affected by the QCD potential and the large width of the top quark. If one aims towards a kinematic determination of the top quark mass and of the tbW coupling in the threshold region these effects must be taken into consideration.

1 Introduction

The study of the top quark production mechanism, the determination of m_t and the investigation of top quark decays are one of the important aims of a future linear electron positron collider. A detailed analysis may lead to the determination of the top quark mass to better than 500 MeV, to a precise value of its electroweak and strong couplings, to a measurement of the QCD potential down to 0.01 fm and may even explore the Yukawa coupling between top quark and Higgs boson. In addition one may derive a variety of limits on couplings outside the framework of the Standard Model, or even discover deviations suggested e.g. in the context of an extended Higgs sector or from residual effective couplings in composite models.

Such a detailed investigation requires the understanding and precise calculation of effects which are characteristic and peculiar for $t\bar{t}$ production, in particular those resulting from the large top quark width of order 1 GeV. This large width leads to the merging of toponium resonances into a fairly smooth but nevertheless still structured

continuum cross section. Interesting structures remain visible for quark masses up to the present limit of about 180 GeV dictated by radiative corrections. Earlier studies were concerned mainly with the behavior of the cross section [1, and references therein], [2–4], or with the momentum distribution of decay products from top quarks in well-separated resonances [5].

Here we shall address the problem of momentum distributions for top quarks and their decay products in the case where individual resonances cannot be distinguished any more. The momentum distribution of top quarks will be related to the Green function for complex energy $E + i\Gamma_t$, which is then obtained through an appropriate numerical analysis.

The following two approximations are inherent in our calculation: non-relativistic treatment and constant decay rate independent of the energy or momentum of the quark. The width of the $t - \bar{t}$ system depends on its intrinsic momentum distribution and on the total energy through the corresponding dependence of both the matrix element squared and the phase space of the decay products. In the decay of a bound top quark the volume of phase space accessible to the decay products is determined by their total energy in their center-of-mass system. This phase space effect tends to reduce the decay rate of bound top quarks relative to free ones [5]. It has been pointed out [6] that the momentum dependent width may modify significantly predictions for the cross section $\sigma(e^+e^- \rightarrow t\bar{t})$ in the threshold region. However, the above mentioned phase space effect is accompanied by the b quark attraction effect. The chromostatic field which attracts the original top quark also attracts the decay b quark, enhancing the amplitude of the b quark near the origin. The overlap between t and b wave functions is larger than for plane wave b quarks, and the matrix element for the decay is therefore increased. It is known that for decays of muons bound in light nuclei analogous effects tend to cancel, and the net result for the decay rate can be well described by the time dilatation effect: the decay rate of a muon momentum component is less in the laboratory frame than in the rest frame of this component [7]. The latter is clearly a relativistic effect. Therefore, the momentum dependent

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width and relativistic corrections should be incorporated simultaneously into calculations or neglected altogether. For the same reason also the P -wave contribution to the differential cross section will be neglected. Furthermore, QCD corrections to decays of on-shell t quarks are known to be of order 10% [8], and no attempt has been made to extend this calculation to the off-shell case. If the ingenuity of our experimental colleagues will lead to high precision experiments substantiating the need for a very accurate calculation of the $t-\bar{t}$ width including all the above mentioned effects, the combination of the various effects can be incorporated into the framework presented in this paper.

W bosons from top quark decay are evidently an ideal probe of $t\bar{t}$ interactions at short distances. In contrast to b -jets they do not suffer from QCD final state interaction. Their distribution reflects the top quark momentum distribution, folded of course with the two-body decay kinematics and smeared by the finite width of the W .

In the following chapter the formalism will be developed that is used to calculate the momentum distribution through the Green function. Section 3 will be devoted to the numerical methods employed in our calculation. In Sect. 4 the results of the calculation of the Green function of the $t-\bar{t}$ system and W energy distributions will be presented. The latter are accessible to experiment and could be used to determine the properties of the top quark as discussed above. Section 6 contains our conclusions.

2 Green function and Lippmann-Schwinger equation

The amplitude for the production of top and antitop quarks with momenta $\mathbf{p}$ and $-\mathbf{p}$, respectively, from a source localized at relative distance $\mathbf{r}=0$ and with oscillatory time dependence

$$j(\mathbf{r}, t) \sim \delta(\mathbf{r}) e^{-iEt} \quad (1)$$

is given by the Green function

$$\mathcal{G}(\mathbf{p}, E) = \int d^3r e^{i\mathbf{p}\cdot\mathbf{r}} G(\mathbf{r}, \mathbf{r}'=0, E) = \left\langle \mathbf{p} \left| \frac{1}{E - H + i\varepsilon} \right| \mathbf{r}=0 \right\rangle \quad (2)$$

with time dependence again $\sim e^{iEt}$. The full Hamiltonian H

$$H = \frac{\mathbf{p}^2}{m_t} + V(\mathbf{r}) \quad (3)$$

is determined by the reduced mass of the $t\bar{t}$ system $\frac{m_t}{2}$ and the potential V to be specified below. The amplitude for the production of unstable t and $\bar{t}$ quarks with decay rate Γ_t decreases proportional to $e^{-\Gamma_t t}$. In the stationary description this behavior can be incorporated through the shift $E \rightarrow \mathcal{E}$, where

$$\mathcal{E} = E + i\Gamma_t. \quad (4)$$

In this case the Green function fulfills the Lippmann-Schwinger equation for complex energies

$$\mathcal{G}(\mathbf{p}, \mathcal{E}) = \mathcal{G}_0(\mathbf{p}, \mathcal{E}) + \mathcal{G}_0(\mathbf{p}, \mathcal{E}) \int \frac{d^3q}{(2\pi)^3} \tilde{V}(\mathbf{p}-\mathbf{q}) \mathcal{G}(\mathbf{q}, \mathcal{E}), \quad (5)$$

where $\tilde{V}(\mathbf{p})$ is the potential in momentum space. The free Hamiltonian that is used to define the Green function $\mathcal{G}_0$ includes the width. Hence, for the $t-\bar{t}$ system

$$\mathcal{G}_0(\mathbf{p}, \mathcal{E}) = \frac{1}{E + i\Gamma_t - \frac{p^2}{m_t}}. \quad (6)$$

With these ingredients one derives a compact expression for the cross section for top quark production in electron positron annihilation (without Z contribution and transverse gluon correction, which can be trivially incorporated):

$$\frac{d\sigma}{d^3p}(\mathbf{p}, E) = \frac{24\pi^2\alpha^2 Q_t^2}{sm_t^2} \Gamma_t |\mathcal{G}(\mathbf{p}, \mathcal{E})|^2. \quad (7)$$

This formula is particularly useful in the case of large width [8]

$$\Gamma_t \approx \frac{G_F m_t^3}{8\sqrt{2}\pi} \left(1 - \frac{m_W^2}{m_t^2}\right)^2 \left(1 + 2\frac{m_W^2}{m_t^2}\right) \left[1 - \frac{2}{3}\frac{\alpha_s}{\pi} f(m_W^2/m_t^2)\right], \quad (8)$$

$$f(y) = \pi^2 + 2\text{Li}_2(y) - 2\text{Li}_2(1-y) + [4y(1-y-2y^2)\ln y + 2(1-y)^2(5+4y)\ln(1-y) - (1-y)(5+9y-6y^2)]/[2(1-y)^2(1+2y)], \quad (9)$$

where one has to consider many nearly overlapping resonances. Expanding $\mathcal{G}(\mathbf{p}, \mathcal{E})$ in terms of solutions of the Schrödinger equation

$$\mathcal{G}(\mathbf{p}, \mathcal{E}) = \sum_n \frac{\tilde{\psi}_n(\mathbf{p}) \psi_n^*(0)}{E - E_n + i\Gamma_t}, \quad (10)$$

and considering the case of well-separated resonances one reproduces the differential momentum distribution from individual resonances

$$\frac{d\sigma}{d^3p}(\mathbf{p}, E) = \frac{24\pi^2\alpha^2 Q_t^2}{sm_t^2} \sum_n |\psi_n(\mathbf{r}=0)|^2 \cdot \frac{\Gamma_t}{(E - E_n)^2 + \Gamma_t^2} |\tilde{\psi}_n(\mathbf{p})|^2. \quad (11)$$

The total cross section as derived from integrating (7)

$$\int \frac{d^3p}{(2\pi)^3} \Gamma_t |\mathcal{G}(\mathbf{p}, \mathcal{E})|^2 = -\text{Im} G(\mathbf{x}=0, \mathbf{x}'=0, \mathcal{E}) \quad (12)$$

coincides with the formula presented in [2]. The LS equation can be solved in analytical form for the Coulomb potential [2]. For the problem under discussion only numerical methods are available which will be described in detail in Sect. 3.

The total cross section, expressed through the Green function with spatial argument $\mathbf{x}=\mathbf{x}'=0$ is most conveniently calculated in coordinate space. The LS integral equation, however, suggests to formulate the problem in momentum space and to solve it directly for $\mathcal{G}(\mathbf{p}, \mathcal{E})$. The perturbative potential, applicable for large momenta, is given by [9]

$$\tilde{V}_{\text{pert}}(Q^2, n_f) = -\frac{16\pi\alpha_{\overline{MS}}(Q^2)}{3Q^2} \left[1 + \left(\frac{31}{3} - \frac{10}{9}n_f\right) \frac{\alpha_{\overline{MS}}(Q^2)}{4\pi}\right]. \quad (13)$$

The renormalization scale μ^2 has been chosen as Q^2 , and n_f refers to the number of massless quarks. Employing standard arguments based on the renormalization group, the Q^2 expansion of $\alpha_{\overline{MS}}(Q^2)$ is given by

$$\alpha_{\overline{MS}}(Q^2) = \frac{4\pi}{b_0 \ln(Q^2/\Lambda_{\overline{MS}}^{(n_f)^2}) + \frac{b_1}{b_0} \ln \ln(Q^2/\Lambda_{\overline{MS}}^{(n_f)^2})}, \quad (14)$$

$$b_0 = 11 - \frac{2}{3}n_f, \quad b_1 = 102 - \frac{38}{3}n_f.$$

For larger distances (smaller Q^2) this potential has to be supplemented by a piece which ensures confinement through a linear rise and properly accounts for $b\bar{b}$ and $c\bar{c}$ spectroscopy. For the numerical studies presented below the two-loop QCD potential, (14), is joined smoothly to a Richardson-like form for intermediate and small momenta:

$$\tilde{V}(\mathbf{p}) = \begin{cases} \tilde{V}_{\text{pert}}(p^2, n_f=5), & \text{if } |\mathbf{p}| > 5 \text{ GeV}, \\ -\frac{4}{3} \frac{48\pi^2}{27} \frac{1}{p^2} \left(\frac{1}{\ln(1+p^2/\Lambda_R^2)} + C \right), & \text{if } |\mathbf{p}| < 5 \text{ GeV}, \end{cases} \quad (15)$$

with $\Lambda_R = 400$ MeV, and C fixed through continuity at the matching point. It is well known that the Richardson [10] potential, and hence the potential (15), reproduces well the spectra of J/ψ and Υ families. Moreover, as a consequence of the large decay rate and the resulting cutoff for large distances, the results depend weakly on the phenomenological part of the potential for realistic top masses and decay rates.

3 Numerical solution of Lippmann-Schwinger equation

The spherically symmetric solution to the Lippmann-Schwinger equation (5), i.e. the S -wave component of the full propagator can be written as

$$\mathcal{G}(p, \mathcal{E}) = \mathcal{G}_0(p, \mathcal{E}) K(p, \mathcal{E}). \quad (16)$$

The vertex function K fulfills the integral equation

$$K(p, \mathcal{E}) = 1 + \int_0^\infty dq \hat{V}(p, q) \mathcal{G}_0(q, \mathcal{E}) K(q, \mathcal{E}), \quad (17)$$

where

$$\hat{V}(p, q) = \frac{1}{4\pi^2} \frac{q}{p} \int_{|p-q|}^{p+q} du u \tilde{V}(u). \quad (18)$$

For potentials that do not vanish at large distances, a fortiori for confining ones, the kernel $\hat{V}(p, q)$ is singular at $p=q$. Even the Coulomb potential $\hat{V}$ exhibits a logarithmic singularity at this point, and for our QCD potential $\hat{V}(p, q) \sim (p-q)^{-2}$ for $|p-q| \rightarrow 0$. Therefore, when solving (17) numerically in momentum space, special consideration will be given to this singularity. Noting that the singularity is associated with low momentum transfers, i.e. large relative distances for $t\bar{t}$ system this difficulty can be circumvented as follows: t and $\bar{t}$, being short lived and slow, cannot penetrate into this region. Therefore, the solution $K(p, \mathcal{E})$ should be insensitive to modifications of

the potential $\tilde{V}(q)$ at very low momenta q . Thus, instead of the potential (15) we consider a family of potentials:

$$\tilde{V}_{\text{cut}}(\mathbf{p}; q_{\text{cut}}) = \begin{cases} \tilde{V}(\mathbf{p}), & \text{if } |\mathbf{p}| > q_{\text{cut}}, \\ -\frac{C_1}{p^2 + q_{\text{cut}}^2} + \tilde{V}_0(q_{\text{cut}}) \delta^{(3)}(\mathbf{p}), & \text{if } |\mathbf{p}| < q_{\text{cut}}. \end{cases} \quad (19)$$

The constant C_1 is fixed by continuity at $p=q_{\text{cut}}$. $\tilde{V}_0(q_{\text{cut}})$ is chosen such that, after Fourier transformation into position space, the potential

$$V_{\text{cut}}(r; q_{\text{cut}}) = \int \frac{d^3q}{(2\pi)^3} e^{-i\mathbf{q}\cdot\mathbf{r}} \tilde{V}_{\text{cut}}(\mathbf{q}; q_{\text{cut}}) \quad (20)$$

does not depend on q_{cut} for small r . Throughout our calculations we derive $\tilde{V}_0(q_{\text{cut}})$ from the condition

$$V_{\text{cut}}(r=1 \text{ GeV}^{-1}) = -\frac{1}{4} \text{ GeV}.$$

This prescription, therefore, provides a regularization of $\tilde{V}(q)$ at $q=0$.

For realistic top masses and widths $K(p, \mathcal{E})$ is independent from q_{cut} for $q_{\text{cut}} < 200$ MeV. To avoid numerical instabilities q_{cut} should be larger than several MeV. Our choice was $q_{\text{cut}} = 50$ MeV. It is noteworthy that for $\alpha_s \sim 0.12$ and $q_{\text{cut}} \lesssim 50$ MeV the resulting potential describes very well (within few tens of MeV) S -wave excitations of J/ψ and Υ .

Let us sketch now our method of solving the integral equation (17). Whenever necessary the potential is regularized as described above, c.f. (19). In order to improve the convergence of the integral containing the unknown function $K(q, \mathcal{E})$ (17) is cast into the following form:

$$[1 - \mathcal{A}(p, \mathcal{E})] K(p, \mathcal{E}) + \int_0^\infty dq \hat{V}(p, q) \mathcal{G}_0(q, \mathcal{E}) \cdot [K(p, \mathcal{E}) - K(q, \mathcal{E})] = 1, \quad (21)$$

where

$$\mathcal{A}(p, \mathcal{E}) = \int_0^\infty dq \hat{V}(p, q) \mathcal{G}_0(q, \mathcal{E}) \quad (22)$$

is a known function which can be evaluated by a numerical integration. The integral can be approximated by a finite sum over discrete values of $q \in Q_N = \{q_1, q_2, \dots, q_N\}$. Considering discrete p from the same set Q_N one approximates the integral equation by a $(N \times N)$ system of linear equations. The later can be solved by standard numerical methods. Accuracy and computer time depend crucially on the choice of Q_N . The integral is split into three parts corresponding to subintervals $(0, a_1)$, (a_1, a_2) and $(0, a_2^{-1})$. In the third part $t=1/q$ defines a new integration variable. The integrals are approximated using the Gauss-Legendre formula:

$$\int_a^b dq f(q) \approx \sum_{i=1}^n w_i f(q_i). \quad (23)$$

The points q_i and weights w_i can be found in standard textbooks on numerical integration [11, 12]. A reasonable choice of subintervals is provided by $a_1 \sim \sqrt{m_t \Gamma_t}$ and $a_2 \sim m_t$.

To study the accuracy and stability of our numerical solutions a number of tests and cross checks has been performed. For the Coulomb potential the results have been compared with known analytical formulae. For QCD the numerical stability of $K(p, \mathcal{E})$ has been studied by changing the number of points N between 100 and 400. The total cross section $\sigma(e^+e^- \rightarrow t\bar{t})$ has been evaluated integrating the differential cross section $\frac{d\sigma}{dp}$ and the results have been compared with those obtained from $\text{Im}\mathcal{G}$ via the optical theorem. The results for the total cross section have been also compared with those obtained by a completely different numerical approach due to Strassler and Peskin [13]. From these studies the numerical accuracy of the results presented in the next chapter is estimated to be better than 1%, and typically of order 10^{-3} .

4 Results

For small top quark masses, small decay widths, and well-separated resonances the Green function exhibits a complicated dependence on $\mathbf{p}$ and $\mathcal{E}$ reflecting the resonance structure of the $t\bar{t}$ system and the large number of nodes for high radial excitations. This is best illustrated for the pure Coulomb potential with $\alpha_s=0.2$, $m_t=150$ GeV and $E=0$, employing for this fictitious case a fairly small width $\Gamma_t=10^{-4}$ GeV (Fig. 1). The oscillatory behavior is cut off for momenta of order $p \sim \sqrt{\Gamma_t m_t}$ in agreement with the physical intuition. For the Coulomb case it was checked numerically that the integrated cross section agrees with the analytical results from [2].

A qualitatively similar behavior is obtained for the realistic case of a top quark mass of 120 GeV with Γ_t from (8) and the QCD potential equation (19). In the region of the 1S resonance one obtains a behavior of the Green function quite similar to the one expected for the wave function of a 1S state (Fig. 2), a broad amplitude with maximal weight around the origin and no node. For $E=0$

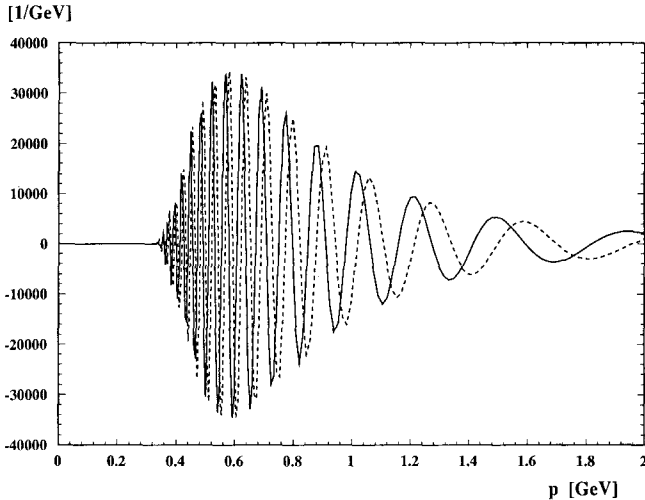


Fig. 1. Green function $\mathcal{G}(p, E + i\Gamma_t)$ for the Coulomb potential with $\alpha_s=0.2$, $m_t=150$, $\Gamma_t=10^{-4}$ and $E=0$ GeV: $\text{Re}\mathcal{G}$ (dashed), $\text{Im}\mathcal{G}$ (solid)

one finds oscillations, a maximum of the amplitude at larger momenta around 10 GeV and damping towards smaller momenta resulting from the nonvanishing width (Fig. 3). For energies above top threshold the amplitudes and in particular the distributions are shifted towards even larger momenta $p \approx \sqrt{Em_t}$. The large momentum tail $\sim p^{-2}$ is clearly visible. This behavior can be deduced already from the free Green function which develops also a large momentum tail despite the fact that it approaches the delta function dictated by phase space in the narrow width approximation.

$$p^2 \Gamma |\mathcal{G}_0(\mathbf{p}, E + i\Gamma)|^2 = \frac{\Gamma p^2}{(E - p^2/m)^2 + \Gamma^2} \rightarrow \begin{cases} \pi m E \delta(E - p^2/m) & \text{for } \Gamma \rightarrow 0, \\ \Gamma m^2/p^2 & \text{for } p^2 \gg mE. \end{cases} \quad (24)$$

This large momentum tail influences the treatment of expectation values as discussed below.

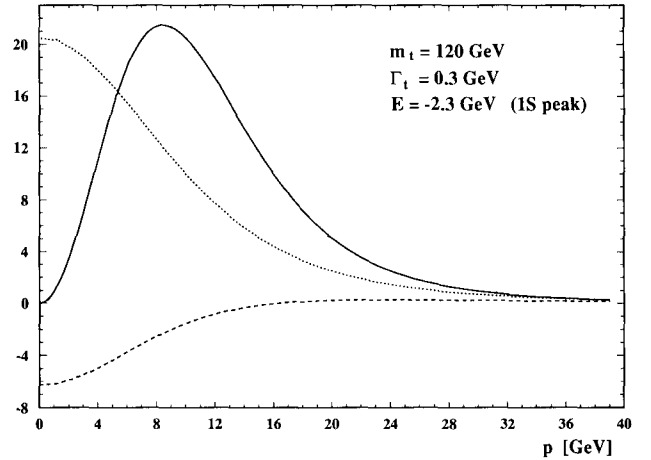


Fig. 2. Green function $\mathcal{G}(p, E + i\Gamma_t)$ for $m_t=120$ GeV and $E=-2.3$ GeV (1S peak): $\text{Re}\mathcal{G}$ [1/GeV] (dashed), $\text{Im}\mathcal{G}$ (dotted), and $|p\mathcal{G}(p)|^2 * 0.002$ (solid)

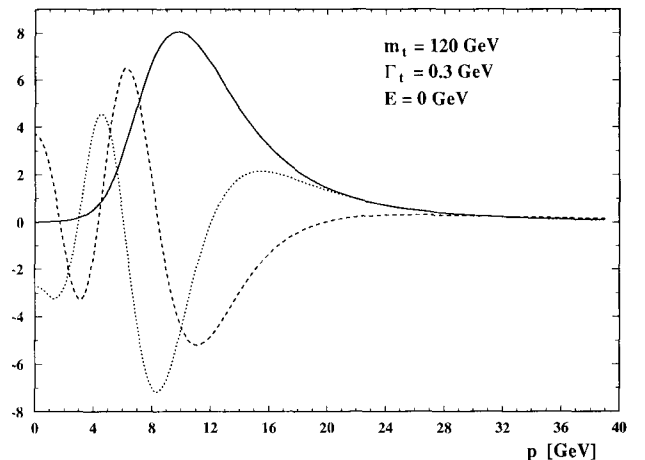


Fig. 3. Green function $\mathcal{G}(p, E + i\Gamma_t)$ for $m_t=120$ GeV and $E=0$: $\text{Re}\mathcal{G}$ [1/GeV] (dashed), $\text{Im}\mathcal{G}$ (dotted), and $|p\mathcal{G}(p)|^2 * 0.002$ (solid)

The square of Green functions are shown in Figs. 4a–c for different top masses (120, 150 and 180 GeV) and for three energies: for E corresponding to the 1S peak positions (–2.3, –2.7 and –2.9 GeV, respectively), for $E=0$, and for $E=2$ GeV. The large decay rate (0.8 and 1.5 GeV

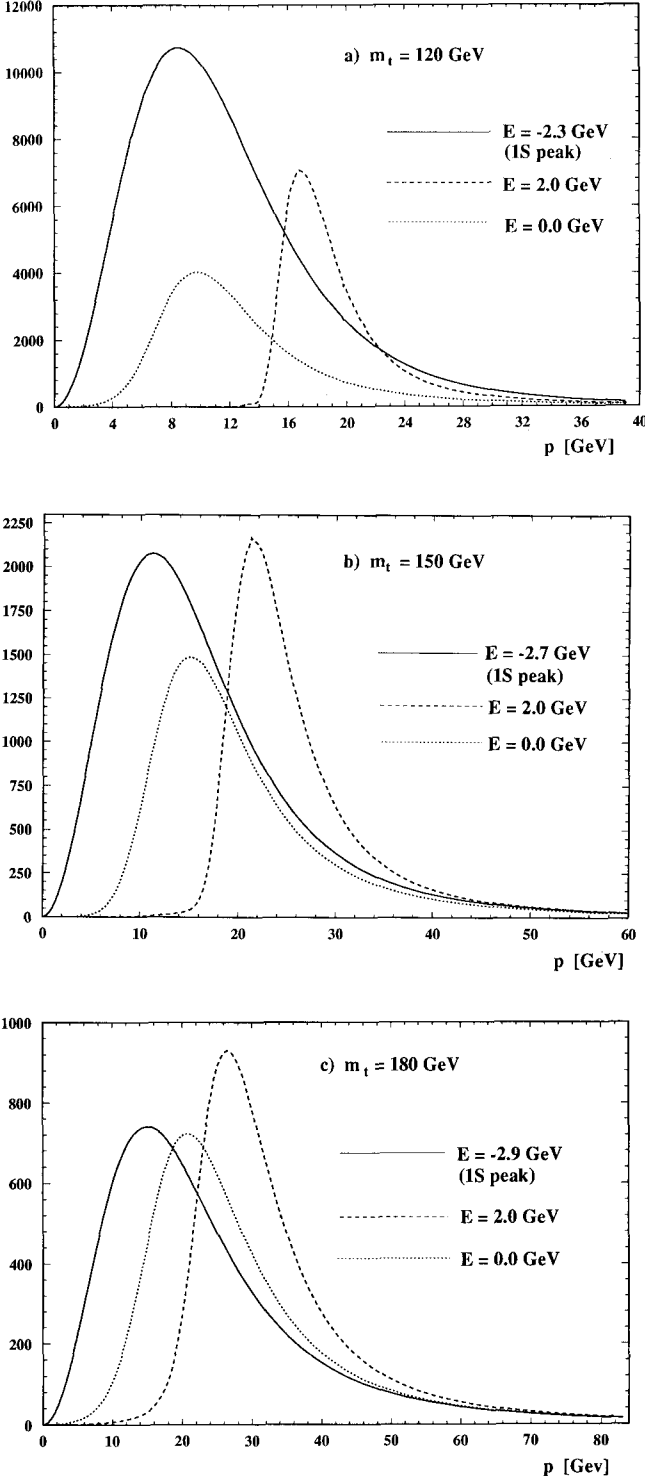


Fig. 4a–c. Green function $|p\mathcal{G}(p, E + i\Gamma_t)|^2$: **a** $m_t = 120$ GeV; **b** $m_t = 150$ GeV; **c** $m_t = 180$ GeV, for E at 1S peak – solid, $E=0$ – dotted, and $E=2$ GeV – dashed line

for $m_t = 150$ and 180 GeV as compared to 0.3 GeV for $m_t = 120$ GeV) damps the oscillations of the Green function and therefore only the momentum distributions are shown. The shift towards large momenta with increasing energy and the narrowing of the distributions is clearly visible. For $E=2$ GeV the behavior is qualitatively already similar to a Breit-Wigner distribution which would result from the Green function without interaction. The relative height just reflects the magnitude of the total cross section. We convinced ourselves that the total cross section obtained from integrating $\frac{d\sigma}{dp}$ coincides numerically with the cross section obtained through the optical theorem from $-\text{Im } G(\mathbf{x}=0, \mathbf{x}'=0, \mathcal{E})$ evaluated in coordinate space (Fig. 5).

A quantity of interest is the average momentum of the top quark and its dependence on m_t , Γ_t and E . For small widths one intuitively expects that $\langle p_t \rangle$ coincides with the average momentum as obtained for the toponium S-wave resonances from the solution of the Schrödinger equation. For larger widths a significant broadening of the momentum distribution is expected and observed. To evaluate the average momentum a cutoff $p_{\max}$ has been introduced to damp the (unphysical) large momentum tail which would lead to a logarithmic divergence for

$$\langle p_t \rangle = \frac{\int_0^{p_{\max}} dp p^3 |\mathcal{G}(p, \mathcal{E})|^2}{\int_0^{p_{\max}} dp p^2 |\mathcal{G}(p, \mathcal{E})|^2} \quad (25)$$

evident already from the free Green function. From the example without interaction it is clear that the variation of the result with the cutoff $p_{\max}$ is about 10%, if the latter is chosen in the physically reasonable range $m_t/2$ to $2m_t$. In the results shown in Fig. 6a $p_{\max} = m_t/2$ is used. The average top quark momentum is large for a tightly bound system ($E = -3$ to -2 GeV), decreases towards the threshold and starts to rise with increasing energy as expected for open top production. The large top decay rate widens

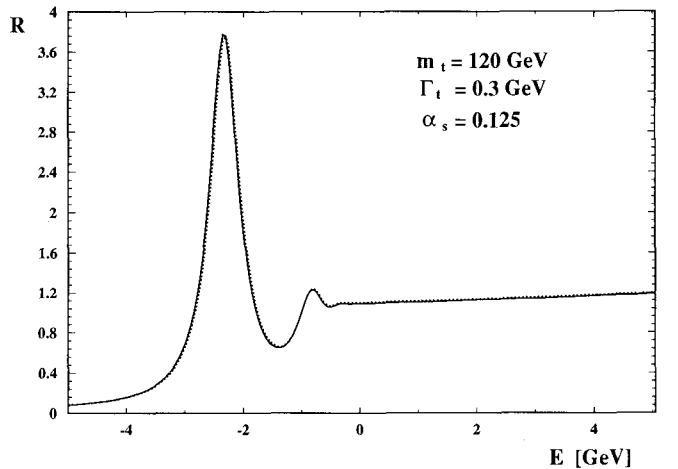


Fig. 5. Comparison of the cross sections $\sigma(e^+e^- \rightarrow t\bar{t})$ for $m_t = 120$ GeV calculated from the differential cross section (7) – solid line and from $\text{Im } G(\mathbf{x}=0, \mathbf{x}'=0, \mathcal{E})$ evaluated in the position space – dotted line

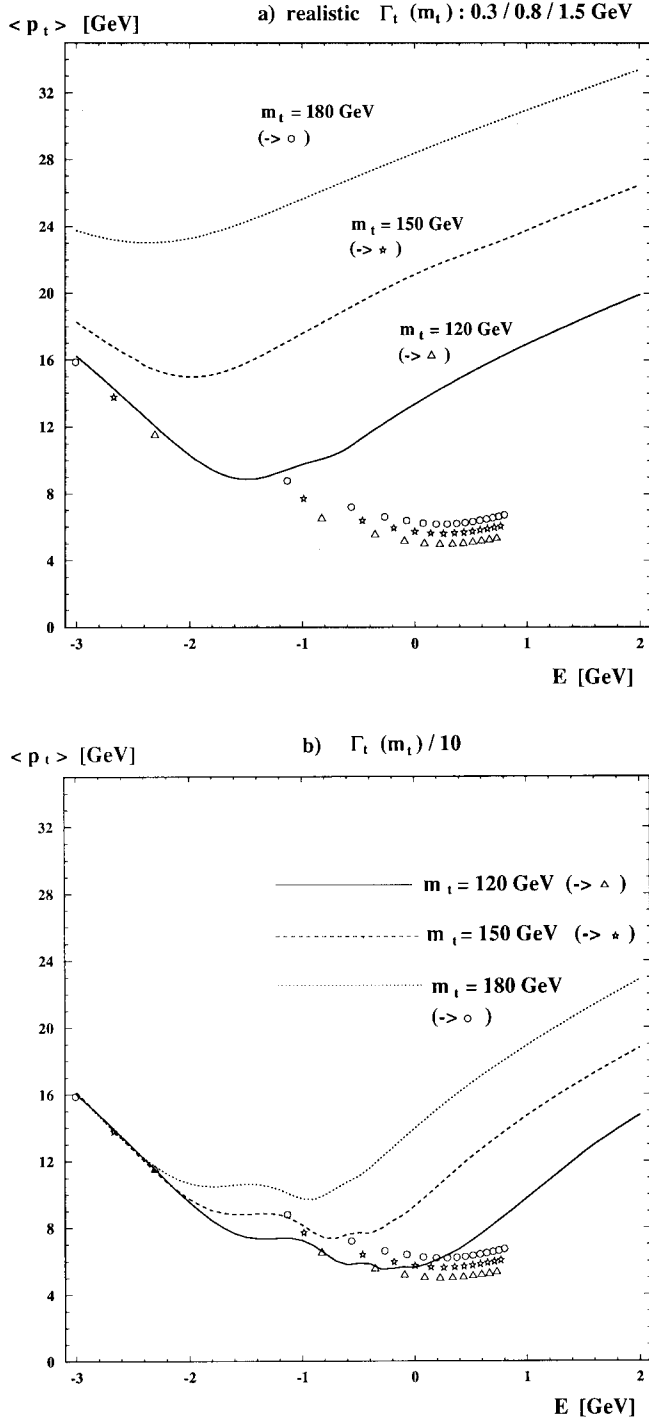


Fig. 6a,b. Energy dependence of $\langle p_t \rangle$ the average t quark momentum for **a** $m_t = 120/150/180$ GeV: solid/dashed/dotted line, respectively, and $\Gamma_t(m_t)$ calculated from (8). Triangles, stars and circles correspond to $\langle p_t \rangle$ for S partial wave bound states and $\Gamma_t = 0$; **b** $\Gamma_t(m_t)/10$

the distribution and from the previous discussion an increase of the average momentum is expected by an additional term of order $\sqrt{m_t \Gamma_t}$. For comparison, the average momenta for the S -wave resonances are indicated at their respective locations. To demonstrate the smooth transition to the resonance description for $\Gamma_t \rightarrow 0$ the cor-

responding calculations were also performed for smaller decay rates ($\Gamma_t/10$). As expected, $\langle p_t \rangle$ gradually approaches the results predicted for individual resonances (Fig. 6b).

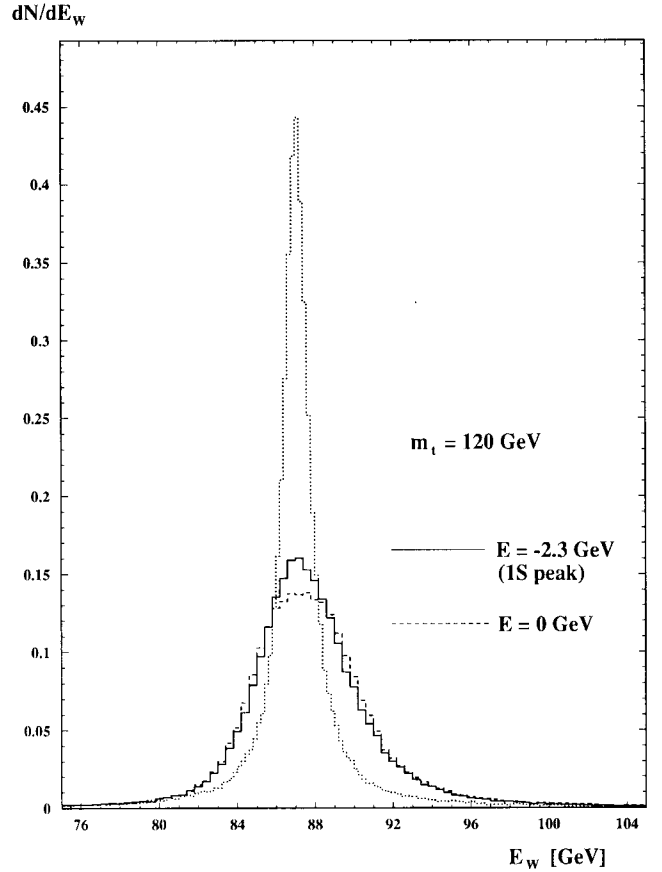


Fig. 7. Comparison of W energy distributions for $m_t = 120$ GeV at $E = -2.3$ GeV (1S peak) - solid, $E = 0$ GeV - dashed, and decays at rest - dotted line

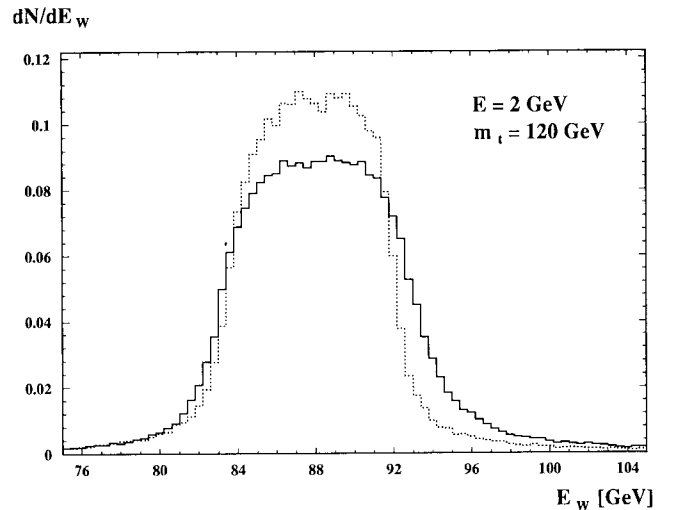


Fig. 8. Comparison of W energy distributions for $m_t = 120$ GeV at $E = 2$ GeV - dashed, and decays of noninteracting t quark with momentum $p = \sqrt{m_t E}$ - dotted line

The top quark distribution per se is evidently not directly accessible. However, the W escapes from the interaction region without final state interaction (in contrast to the b jet). It is therefore an ideal probe of this distribution

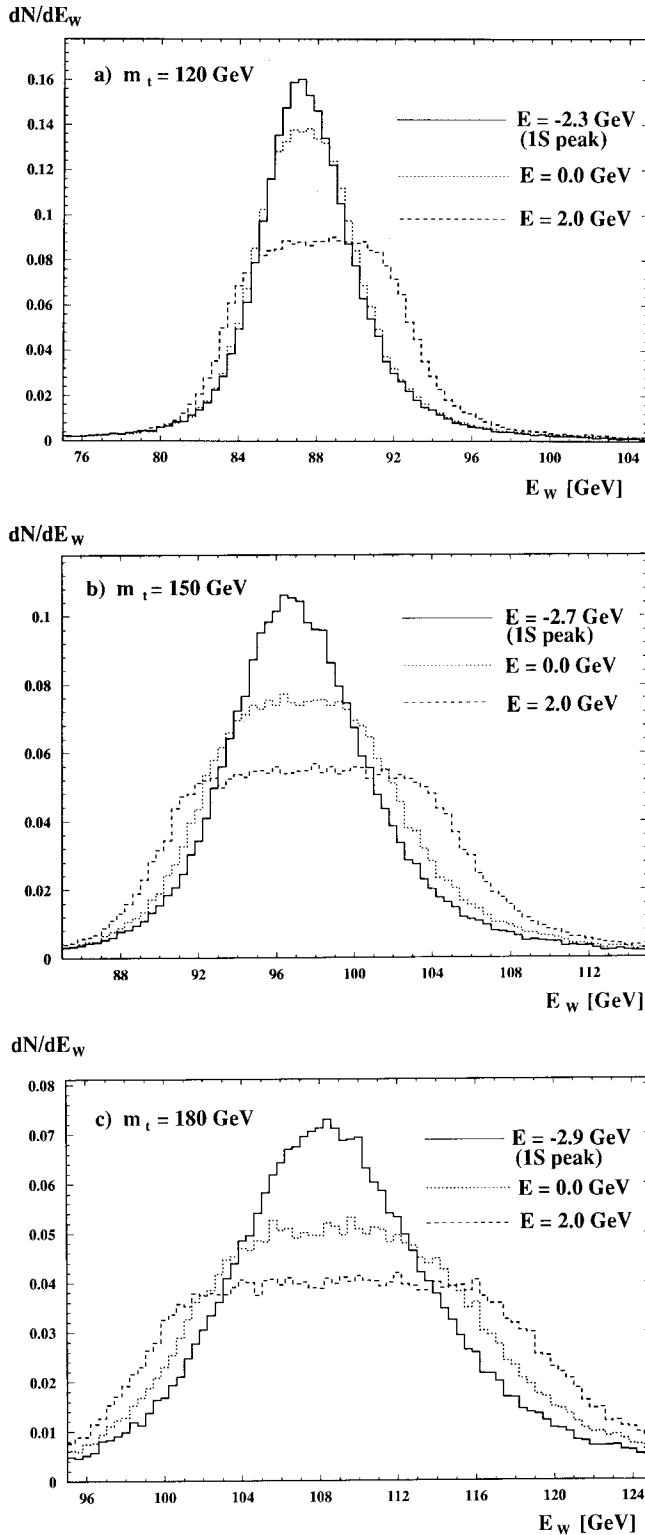


Fig. 9a–c. W energy distributions for **a** $m_t = 120$ GeV; **b** $m_t = 150$ GeV; and **c** $m_t = 180$ GeV: at 1S peak ($E = -2.3/-2.7/-2.9$ GeV, respectively)–solid, $E = 0$ –dotted, and $E = 2$ GeV–dashed line

and hence of the potential at small distances. Conversely, if one wants to employ the W momentum distribution to reconstruct the top quark mass or to determine the tbW coupling the smearing from this “Fermi” motion must be taken into consideration.

Due to the momentum distribution of the decaying t and $\bar{t}$ the resulting W energy spectrum is much broader. In Fig. 7 the spectra for $m_t = 120$ GeV at $E = -2.3$ and $E = 0$ GeV are compared with the spectrum resulting from decays at rest. In Fig. 8 the spectrum for $E = 2$ GeV is compared with the W spectrum from the decay of noninteracting top quark with momentum $p = \sqrt{Em_t}$. For the W width $\Gamma_W = 2.2$ GeV is used throughout.

The W energy distributions for top masses of 120, 150 and 180 GeV are shown in Fig. 9a–c. It is evident that the resulting distributions are sensitive to E and allow the determination of the top mass independent of the total cross section. Moreover, distortions of energy distributions of W s and b quarks due to strong interactions have to be taken into account if one aims towards a precise kinematical determination of m_t in the threshold region as suggested in [14].

The top and W distributions reflect also the strength of the interaction. This is demonstrated in Fig. 10.

5 Summary

The Green function formalism for the description of $t\bar{t}$ production in e^+e^- annihilation close to threshold has been extended from the calculation of the total cross section to the prediction of momentum distributions of top quarks and their decay products. The resulting distributions depend significantly on the width of the top quark and the strength of the QCD potential. If one aims towards a kinematic determination of the top quark mass and of the tbW coupling in the threshold region these effects must be taken into consideration.

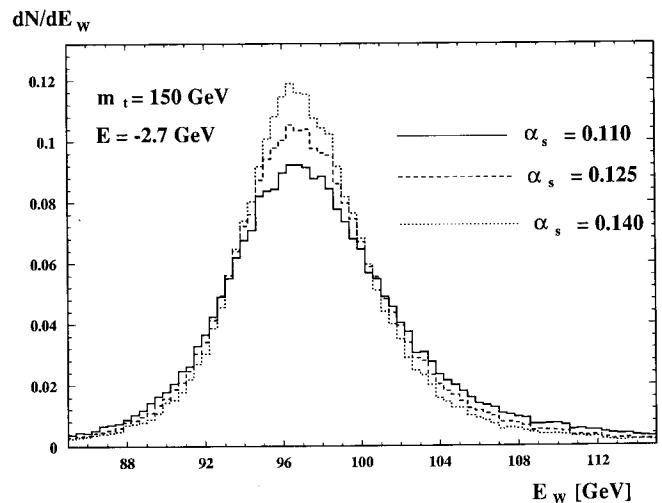


Fig. 10. W energy distributions for $m_t = 150$ GeV at $E = -2.7$ GeV (1S peak) and different values of $\alpha_s(M_Z)$: 0.11–solid, 0.125–dashed, and 0.14–dotted line

Note added. After completion of this work we received a preprint by Y. Sumino et al. [15] where momentum distributions of top quarks are also calculated. In contrast to our work the Green function is evaluated in coordinate space. A momentum dependent width for the $t-\bar{t}$ system is introduced which, however, does not include effects due to the potential energy of b quark in the chromostatic field. Since our calculations have been performed for constant width the results of the works are only in qualitative agreement.

References

1. J.H. Kühn, P. Zerwas: Phys. Rep. 167 (1988) 321; J. H. Kühn, P. Zerwas; Heavy Flavours in High Energy Electron-Positron Collisions, in: A. Buras, M. Lindner (eds.), Review volume on heavy flavours. Singapore: World Scientific 1992
2. V.S. Fadin, V.A. Khoze: JETP Lett. 46 (1987) 525; Sov. J. Nucl. Phys. 48 (1988) 309
3. W. Kwong: Phys. Rev. D 43 (1991) 1488
4. J.M. Strassler, M.E. Peskin: Phys. Rev. D 43 (1991) 1500
5. M. Jezabek, J.H. Kühn: Phys. Lett. B 207 (1988) 91
6. K. Fujii: talk presented at the Workshop on Physics and Experiments at Linear Colliders, Saariselkä, Finland 1991
7. R.W. Huff: Ann. Phys. 16 (1961) 288
8. M. Jezabek, J.H. Kühn: Nucl. Phys. B 314 (1989) 1
9. W. Fischler: Nucl. Phys. B 129 (1977) 157; A Billoire: Phys. Lett. B 92 (1980) 343
10. J.L. Richardson: Phys. Lett. B 82 (1979) 272
11. M. Abramowitz, I.A. Stegun (eds.): Handbook of mathematical functions. Washington D.C.: US National Bureau of Standards, 1966
12. W.H. Press, B.P. Flannery, S.A. Teukolsky, W.T. Vetterling: Numerical recipes. The art of scientific computing. Cambridge: Cambridge University Press 1986
13. T. Teubner: Diploma thesis (unpublished)
14. G. Bagliesi et al.: Top Quark Physics – Experimental Aspects, preprint CERN-PPE/92-05
15. Y. Sumino et al.: Top Quark Pair Production Near Threshold, Univ. of Tokyo preprint UT-594